

Third Assignment

1. Generate an object in **R** (call it whatever you want to name it) with 100 random draws from a standard normal distribution. Using this object, calculate and provide:

- A histogram of the object
- The mean
- The standard deviation
- The standard error of the mean
- The 95% confidence interval

2. Use the following data to conduct a difference of means test:

Female representation	Number of observations	Mean Military GDP	SD Military GDP
Low	9	2.00	1.32
High	11	2.64	1.75
Combined	20	2.35	1.57

(a) Calculate the t-statistic.

(b) Calculate the degrees of freedom.

(c) Compare the t-statistic to the critical value for $p = 0.05$ for the corresponding degrees of freedom.

*** If you want to check your answers for #2 in R, you can paste the following code to create the dataset **countrydata** and conduct a t-test:

```
nation <- c("Angola", "Argentina", "Belize", "Burundi", "Chad", "Cote
d'Ivoire", "Cyprus", "Guatemala", "India", "Kenya", "South
Korea", "Lebanon", "Malaysia", "Romania", "Sierra
Leone", "Suriname", "Swaziland", "Thailand", "United States", "Zambia")
#
womenparliament <- c(16, 34, 3, 18, 6, 9, 11, 8, 8, 7, 13, 2, 9, 11, 15, 18, 11, 9, 14, 12)
militarygdp <- c(4, 1, 1, 7, 1, 2, 2, 1, 3, 2, 2, 5, 2, 2, 2, 1, 2, 1, 4, 2)
femalerep <- NA
femalerep[womenparliament < 10] <- "Low"
femalerep[womenparliament > 10] <- "High"
femalerep <- as.factor(femalerep)
#
countrydata <- data.frame(nation, womenparliament, militarygdp, femalerep)
```